PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials
AUTHORS	Ferreira-Valente, Alexandra; Jarego, Margarida; Queiroz-Garcia, Inês; Pimenta, Filipa; Costa, Rui; Day, Melissa; Pais-Ribeiro, José; Jensen, Mark

VERSION 1 – REVIEW

REVIEWER	Marta Illueca Berkeley Divinity School, Yale Program Medicine Spirituality and Religion
REVIEW RETURNED	14-Feb-2021

GENERAL COMMENTS	Dear Authors, This protocol addresses a very important emerging area of interest within the space of spirituality and health. As a researcher in the same area of interest, I remain available for review of further revisions. Please consider carefully my recommendation to not mix individual and group based prayer types. Detailed comments are attached. It is my opinion that by giving consideration to the focusing the choice of prayer type as an intervention, and avoiding the mixing of two different interventions as discussed above, this protocol could help expand the existing knowledge base around the use of prayer as a potential intervention in the management of pain. Best wishes! Reviewer's comment for Manuscript ID bmjopen-2020-047580 This protocol addresses a very important emerging area of interest within the space of spirituality and health. The authors propose a systematic review of the effects of any kind of prayer on a variety of pain-related outcomes. Although the methodology as laid out complies with the PRISMA protocol and CONSORT risk of bias assessment, the chosen area of study, prayer and pain is an area already explored in earlier spirituality and health research, and the authors may want to reconsider the focus their study in a more stringent way so as to build upon, rather than repeat or revisit areas of research that have fallen into disfavor in the clinical research community. The latter includes all manner of distant intercessory group prayer where the person or patient at the center of the prayer intention is not directly participatory in such prayer practices. Published systematic reviews and analysis of studies of the genre of prayer have noted the The lack of a discernible effect on health-related outcomes and the inability to control for the prayer style, frequency and content of group prayer or distant intercessors: Masters, K. S., Spielmans, G. I., & Goodson, J. T. (2006). Are there demonstrable effects of distant intercessory prayer? A meta-analytic

	review. Annals of behavioral medicine, 32(1), 21-26. Masters, K. S., & Spielmans, G. I. (2007). Prayer and health: Review, meta-analysis, and research agenda. Journal of behavioral medicine, 30(4), 329-338. Vincent, Savannah (2016) "Livin' on a Prayer: An Analysis of Intercessory Prayer Studies, "Dialogue & Nexus: Vol. 3 , Article 13. Available at: https://digitalcommons.acu.edu/dialogue/vol3/iss1/13 In addition, distant intercessory prayer is not considered as a scripturally based practice in any of the major world religions so that a consensus has been reached to recommend refocusing research efforts and steer away from studies where the intervention, i.e. prayer is not well defined nor well controlled for. It would be advisable for the research team to consider a consultation with a theologian or pastoral care expert that may further help clarify the notable differences between individual and group or community prayer types. In contrast, recent research (see Illueca & Doolittle, 2020) refocused the study of prayer as a potential therapeutic intervention for pain, based on the type of prayers that are personal (by the person or with the person, at the bedside) with well-defined types of prayer and well controlled clinical settings that were the subject of randomized controlled trials that could provide quantitative data. This data was synthesized in a qualitative summary that described the type of patient profile amenable to prayer interventions, and the types of personal prayer that were found to be beneficial in various acute, surgical or chronic pain settings. Illueca & Doolittle (2020) coined the term "prayer-primed patient" to describe the emerging profile of such patients as informed by their review of the literature. This latter point is misconstrued on page 7, line 4 and should be revised. Based on the above, it is my impression than In its present form this protocol requires a major revision in the following areas (numbered according to the Review Checklist): 1- Study objective is not clearly defined due to a conflation of prayer techniques which are not comparable either in style, content or mode of implementation. The objective of the present study is very similar to the Illueca and Doolittle (2020) systematic review discussed by the protocol authors (pages 6-7) except for the inclusion of all kinds of prayer, which we strongly recommend against. To include both personal and distant intercessory prayer types is akin to mixing group therapy techniques with individual psychotherapy, or similar to studying different types of drugs given at different doses, at varying intervals, with different mechanisms of action and expecting to obtain a meaningful conclusion as to their efficacy as if they were all equivalent. It is advisable for the authors to avoid mixing all types of prayer and if they remain intent on doing so, a stronger rationale that addresses this vital conceptual challenge needs to be put forth. Perhaps a consideration could be given to either performing two separate meta-analysis per type of prayer (personal and community/group based, distant or on site) or to consider refocusing their study on only one kind of prayer. In this respect, another potentially important consideration is to expand on the earlier findings by Illueca and

	Doolittle (2020) by including papers written on all languages (not just the few chosen by the research team) and if the findings allow, to then perform a more robust meta-analytic exercise. Illueca and Doolittle (2020) noted in their study limitations, the high heterogeneity of the data and the limited numbers of studies that met criteria which did not support the performing of a meta-analysis. For this current protocol, the authors may want to reconsider the already reported challenges of previous research. Of note, the authors of this protocol have incorrectly assumed that the Illueca & Doolittle (2020) systematic review focused on and summarized "the benefits of active prayer to a deity as adjunctive therapy for pain") and that this study "focused specifically on the effects of prayer to a deity" (page 6, line12, 15-18). Both of these assumptions are incorrect, and most be revised. The data around active versus passive prayer types (Meints et al, 2018) and the data around prayer to a deity in religious versus non-religious participants (see Jegindo et al, 2013) were findings noted in the systematic review results and not the focus of the study. A careful revision of the results of the Illueca & Doolittle study is recommended, since very specific personal prayer types were identified that seemed beneficial for various types of pain and a correction of the statements noted above is necessary. In addition, I would advise the authors to stay away from the grey literature, with the only exception of preprints, unless there is a clear rationale for an RCT to remain unpublished. The lack of peer review of any data included would diminish the objectivity and value of their findings. I also would recommend a revision of the nine original studies included by Illueca & Doolittle (2020). The introduction of the present protocol could thus be more complete by including the following two studies in Muslim patients reported in the literature: Rahman, F. S., Yahya, N., Din, N. M. M., Izaham, A., & Mat, W. R. W. (2018). The comparative effects of listening to prayer recitation and music therapy intraoperatively on postoperative pain. International Medical Journal Malaysia, 17(2), 107-115. Hassanpour-Dehkordi, A., Fatehi, D. & Solati, K. (2016). Analgesic plus prayer versus analgesic alone. effect of prayer on intensity of postoperative pain, anxiety and physiological indices in surgical patients. A randomized clinical trial. Heroin Addiction and Related Clinical Problems,18(6),13-20. Regarding the search strategy and definitions of variable: - Previous researchers (see above) have noted in their publications the challenges of using very broad search terms and a strategy for streamlining the terms and also for detailing papers to be excluded needs additional clarification - It is advisable to also consider the International Association for the Study of Pain's (IASP) definition of pain for consistent with published literature, as this protocol will explore the emotional and spiritual aspects of pain - Likewise, for consistency with previous systematic reviews, please clearly define the types of prayer to be studied and provide at least one clear definition of prayer, including what is meant by individual versus group prayers which are ver distinct practices from each other as are meditation, yoga, kung fu, reiki, etc.

	3- Study design is not appropriate to answer the research question. Please see refer to the introductory comments above. The chosen area of study, prayer and pain is an area already explored in earlier spirituality and health research, and the authors may want to reconsider the focus their study in a more stringent way so as to build upon rather than repeat or revisit areas of research that have fallen into disfavor in the clinical research community. Therefore, the main flaw of the study design, which is simple to resolve, is the intent to mix two very different types of prayers as interventions which would not be appropriate to answer the research question in a way that would provide a meaningful interpretation of the data. Previous research has largely indicated the inability to properly control prayers used in the community separate from the person at the center of the prayer efforts. It would be a mistake to conflate distant prayer by third parties with personal prayer types and it would be a lost opportunity to provide meaningful data and obscure a potential real relationship between one type of well-defined prayer and the chosen pain-related outcomes. 6- Are the outcomes clearly defined ? As stated above, it is highly improbable to be able to interpret data that is based on a mixture of prayer types. - Research that uses “prayer” as a catch-all umbrella term representing a potential therapeutic intervention can no longer be approached in generic terms. Group based prayers (distant and intercessory) and personal prayer are as distinct as group therapy versus individual therapy. To consider “prayer” as a mixture of techniques and styles severely compromises an accurate interpretation of data, analogous to including both antibiotics and anti-hypertensive medications in the same study in order to prove that a pharmacological intervention can be accurately evaluated without attention to the type, dose and administration routes and frequency of those drugs. - Based on existing research around distant intercessory prayer, there is now a consensus among researchers that the contradictory data on distant intercessory prayer does not justify further research. A major flaw of studies on group based prayers is the inability to control it as a variable where different people pray in their own style for varying durations and with different levels of emotional involvement with the prayer intentions. 6- Are the study limitations discussed adequately? Yes, but there is room for improvement based on the published literature. - Previous research has shown the high degree of heterogeneity of the data related to prayer. The literature shows that methodology of the studies of spirituality, prayer, health and pain is mostly inconsistent between studies described. This substantive limitation should be addressed. - There is a serious lack of measurement tools and validated questionnaires to characterize prayer used by pain patients and most studies do not clearly define the meaning of prayer. There is an added level of heterogeneity by the use of varying types of scales, and the also the use of mixed methodology. - Due to the reported lack of uniformity in research methods it is highly desirable that future study designs like the one proposed in

	this protocol really consider ways in which to provide consistency of definitions of variables especially those pertaining to prayer types. It is my opinion that by giving consideration to the focusing the choice of prayer type as an intervention, and avoiding the mixing of two different interventions as discussed above, this protocol could help expand the existing knowledge base around the use of prayer as a potential intervention in the management of pain.

REVIEWER	Bonnie Huang Hall University of California San Francisco
REVIEW RETURNED	16-Feb-2021

GENERAL COMMENTS	I am not on expert on conducting systematic reviews, nor meta-analysis. I leave that for other experts to comment on. The question the paper tries to answer is the effect of prayer as an intervention for all types of pain. However, the title implies the question is whether prayer is effective for pain management. It may help to have predefined notions of what is effective pain control - given the title implies that the question about to be answered is whether prayer is EFFECTive for pain control? Consequently, possible definitions for what is effectiveness (ie commonly accepted definitions of effectiveness) include 30% reduction in quantitative pain scores, very effective often is 50% reduction in pain scores.). Another comment : the data being gathered from each study may be too generic. Just simply "type of prayer" and "frequency of prayer". It may not be able to show conclusive results since the subject's attitude toward the religion and religious/spiritual beliefs itself may be factors that may influence whether pray is effective. Since the studies are randomized, it may be useful to have other data points added that characterize the subjects stance on religion such as what religious /spiritual affiliation the subjects had, or what type of practitioner they are of that religion/spiritual affiliation (practicing, no religious affiliation, negative attitude towards deity vs positive attitude towards deity, are they self proclaimed as "spiritual" or not, etc. Perhaps a clarification of "type of prayer" can be made. Do the authors refer to group prayer, individual prayer, asking for prayer, or does the author refer to active prayer vs passive prayer. i.e. Type of prayer can be broken down into separate data categories, such as who is praying, how are they praying, etc. it will be hard to reproduce the study if the parameters of "type of prayer" are not specified.

REVIEWER	Alisa Johnson University of Florida
REVIEW RETURNED	28-Feb-2021

GENERAL COMMENTS	I commend you for taking on this topic and I look forward to reading about the study when it is completed. I have no suggestions for revisions. Well done!

VERSION 1 – AUTHOR RESPONSE

REVIEWERS' COMMENTS	AUTHOR RESPONSE
Reviewer: 1 **Dr. Marta Illueca, Berkeley Divinity School** **This protocol addresses a very important emerging area of interest within the space of spirituality and health. As a researcher in the same area of interest, I remain available for review of further revisions.**	We thank the reviewer for these kind comments.
Reviewer: 1 **Please consider carefully my recommendation to not mix individual and group-based prayer types. Detailed comments are attached. It is my opinion that by giving consideration to the focusing the choice of prayer type as an intervention, and avoiding the mixing of two different interventions as discussed above, this protocol could help expand the existing knowledge base around the use of prayer as a potential intervention in the management of pain. Best wishes!**	Thank you very much for your encouragement as well as your helpful comments and suggestions. We agree with your points, and have revised the review to include only papers that present findings from studies evaluating *participatory* individual and group-based prayer types and analyze them separately.
Reviewer: 1 **This protocol addresses a very important emerging area of interest within the space of spirituality and health. The authors propose a systematic review of the effects of any kind of prayer on a variety of pain-related outcomes. Although the methodology as laid out complies with the PRISMA protocol and CONSORT risk of bias assessment, the chosen area of study, prayer and pain is an area already explored in earlier spirituality and health research, and the authors may want to reconsider the focus of their study in a more stringent way so as to build upon, rather than repeat or revisit areas of research that have fallen into disfavor in the clinical research community. The latter includes all manner of distant intercessory group prayer where the person or patient at the center of the prayer intention is not directly participatory in such prayer practices. Published systematic reviews and analysis of studies of the genre of prayer have noted the lack of a discernible effect on health-related outcomes and the inability to control for the prayer style, frequency and content of group prayer or distant intercessors:** **Masters, K. S., Spielmans, G. I., & Goodson, J. T. (2006). Are there demonstrable effects of distant intercessory prayer? A meta-**	We agree that there is a clear distinction between *participatory* (including, individual, dyadic [i.e., two people praying together], and group-based [i.e., more than two people praying together]) prayer, where the targeted beneficiary of the prayer is one or more of the people doing the praying, and distant *intercessory* prayer, where the targeted beneficiary of the praying is someone other than the person or people doing the praying. We also thank the reviewer's suggestion of consulting with a theologian or pastoral care expert to clarify the differences between the different types of prayer, which the research team has done in addition to carefully reviewing previous literature of theoretical and theological models and taxonomies of prayer. We note that not all dyadic or group-based prayer is intercessory. As highlighted by Pérez et al. (2011, doi: 10.1007/s10865-011-9333-9) "prayer is a complex multidimensional behavior and there is no dominant theoretical model or consistent approach to measuring prayer types". While there seems to be a consensus in regards to the definition of prayer (a form of "spiritual communication with God or an object of worship"; Anderson & Nunnelley, 2016, doi: 10.1080/00325481.2016.1209962) there is as of yet no consensus taxonomy for the different types of prayer, as highlighted by Pérez et al. Also, as noted by Anderson and Nunnelley (2016, doi: 10.1080/00325481.2016.1209962) and by Baesler

analytic review. Annals of behavioral medicine, 32(1), 21-26. **Masters, K. S., & Spielmans, G. I. (2007). Prayer and health: Review, meta-analysis, and research agenda. Journal of behavioral medicine, 30(4), 329-338.** **Vincent, Savannah (2016) "Livin' on a Prayer: An Analysis of Intercessory Prayer Studies, "Dialogue & Nexus: Vol. 3 , Article 13. Available at: https://digitalcommons.acu.edu/dialogue/vol3/iss1/13** **In addition, distant intercessory prayer is not considered as a scripturally based practice in any of the major world religions so that a consensus has been reached to recommend refocusing research efforts and steer away from studies where the intervention, i.e. prayer is not well defined nor well controlled for. It would be advisable for the research team to consider a consultation with a theologian or pastoral care expert that may further help clarify the notable differences between individual and group or community prayer types.**	(1999, https://digitalcommons.odu.edu/communication_fac_pubs/8), prayer can be private (an individual praying alone) or communal (an individual prays with another person or group of people). Prayer can also differ with respect to content, including prayers of petition, thanksgiving, adoration, confession, reception, colloquial, meditative, or ritualistic. A prayer of petition includes a request to the deity or object of prayer towards oneself or towards another person or group of people. In this last case, the petition prayer is also called a remote or distant intercessory prayer. Thus, as these authors note, the number of individuals involved in the praying (i.e., just one individual or more than one individual) does not determine in and of itself the content of the prayer. As such, so dyadic, group, shared or communal prayer may or may not be a remote or distant intercessory prayer. There are many examples, within different religious denominations, of group participatory prayer which are not intercessory. We revised the paper to make these issues and distinctions clearer, and note that previous reviews have found that the evidence for the effects of intercessory prayer are weak (thank you for the citations, which we now cite in the revision), and the focus of this review is on individual and communal (i.e., dyadic or group-based) participatory prayer only, excluding all types of intercessory prayer, as suggested. We also planned to analyze separately the results with respect to individual versus communal participatory prayer.
Reviewer: 1 **In contrast, recent research (see Illueca & Doolittle, 2020) refocused the study of prayer as a potential therapeutic intervention for pain, based on the type of prayers that are personal (by the person or with the person, at the bedside) with well-defined types of prayer and well controlled clinical settings that were the subject of randomized controlled trials that could provide quantitative data.** **This data was synthesized in a qualitative summary that described the type of patient profile amenable to prayer interventions, and the types of personal prayer that were found to be beneficial in various acute, surgical or chronic pain settings. Illueca & Doolittle (2020) coined the term "prayer-primed patient" to describe the emerging profile of such patients as informed by their review of the literature. This latter point is misconstrued on page 7, line 4 and should**	Thank you for noting that we had misconstrued this point in the paper. We revised the text in the Introduction to include these ideas.

be revised.	
Reviewer: 1 **Based on the above, it is my impression than in its present form this protocol requires a major revision in the following areas (numbered according to the Review Checklist):** **1- Study objective is not clearly defined due to a conflation of prayer techniques which are not comparable either in style, content or mode of implementation.** **The objective of the present study is very similar to the Illueca and Doolittle (2020) systematic review discussed by the protocol authors (pages 6-7) except for the inclusion of all kinds of prayer, which we strongly recommend against. To include both personal and distant intercessory prayer types is akin to mixing group therapy techniques with individual psychotherapy, or similar to studying different types of drugs given at different doses, at varying intervals, with different mechanisms of action and expecting to obtain a meaningful conclusion as to their efficacy as if they were all equivalent.** **It is advisable for the authors to avoid mixing all types of prayer and if they remain intent on doing so, a stronger rationale that addresses this vital conceptual challenge needs to be put forth.** **Perhaps a consideration could be given to either performing two separate meta-analysis per type of prayer (personal and community/group based, distant or on site) or to consider refocusing their study on only one kind of prayer.**	We agree with the Reviewer's recommendation to clarify the aims of the planned Systematic Review, limiting this review to participatory prayer, as described above. The objective of the present study is therefore somewhat broader than Illueca and Doolittle's (2020) review. Furthermore, as highlighted in the last paragraphs of the Introduction, we believe that the planned Systematic Review will build upon Illueca and Doolittle's (2020) work, such that it is conducted using the current state-of-the science standards and recommendations for rigorous systematic reviews. Thus, to build on this previous work, we (1) did not restrict the literature searches to articles published in English only (we included six languages), (2) expanded the dates of publication of the potential articles (we included all dates up to the present), (3) expanded the sources to more databases (we plan to search nine databases, including the grey literature, as recommended by the "Cochrane Handbook for Systematic Reviews of Interventions" (see, for example, section 4.3.5. of this manual for the recommendation to search the grey literature), and (4) included papers that study communal participatory prayer. Thus, we believe that the planned Systematic Review will be more comprehensive and will reduce the risk of publication and selection bias, relative to previous reviews. In addition, and also as recommended by COCHRANE, the planned Systematic Review will use a standardized method for performing a qualitative analysis and for performing the evaluation of the methodological quality of the articles selected for review (see Method section). We agree that it would be useful to exclude distant intercessory prayer, per our previous responses. We also agree that individual and communal-based participatory prayer are different and have the potential to have different effects. We revised the protocol of the Systematic Review to plan to evaluate these prayer types separately, and discuss the implications of the distinctions among these prayer types by future researchers.
Reviewer: 1 **In this respect, another potentially important consideration is to expand on the earlier findings by Illueca and Doolittle (2020) by including papers written on all languages (not just the few chosen by the research team) and if the findings allow, to then perform a more robust meta-analytic exercise. Illueca and Doolittle (2020) noted in their study limitations, the high**	We agree with the reviewer that including manuscripts published in *all* languages would make the Systematic Review even more robust than limiting the Systematic Review to studies published in 6 languages (English, German, French, Spanish, Italian and Portuguese). However, as we explain in our originally submitted manuscript, the research team does not possess the resources to review and include manuscripts in

heterogeneity of the data and the limited numbers of studies that met criteria which did not support the performing of a meta-analysis.	all languages (i.e., there are over 6,000 languages spoken in the world). Nevertheless, by including manuscripts published in 6 languages – the ones spoken by the research team members – the planned review has the potential to be more comprehensive than if it were restricted to papers published in English. This will also potentially reduce publication and selection bias further. We note that three of these languages are within the top-10 most spoken languages in the world. Spanish is the second most spoken language in the world, followed by English (the third most spoken language). Portuguese is the seventh most spoken language in the world. By including manuscripts published in 5 languages other than English, the planned Systematic Review has the potential to include manuscripts that were not identified and included in previous reviews. Furthermore, the specific inclusion criterion for the planned review and those of other previous reviews differ. Thus, for example, it is possible that manuscripts included in the planned Systematic review may differ from those included by Illueca and Doolittle (2020). We are therefore unable to determine at this point if it will be possible to perform meta-analysis with at least two manuscripts with low heterogeneity of the data.
Reviewer: 1 **For this current protocol, the authors may want to reconsider the already reported challenges of previous research.**	We were not sure what the reviewer is specifically referring to here (i.e., by "challenges" does the reviewer mean "weaknesses" of previous research, the challenges of conducting rigorous research in this area, or something else?). Thus, we did not revise the text to address this specific point, and hope that the revisions made in response to other points address it.
Reviewer: 1 **Of note, the authors of this protocol have incorrectly assumed that the Illueca & Doolittle (2020) systematic review focused on and summarized "the benefits of active prayer to a deity as adjunctive therapy for pain") and that this study "focused specifically on the effects of prayer to a deity" (page 6, line12, 15-18). Both of these assumptions are incorrect, and most be revised.** **The data around active versus passive prayer types (Meints et al, 2018) and the data around prayer to a deity in religious versus non-religious participants (see Jegindo et al, 2013) were findings noted in the systematic review results and not the focus of the study. A careful revision of the results of the**	We thank the reviewer for noticing these inconsistencies. We revised the manuscript in order to more accurately summarize the purpose and results of Illueca & Doolittle (2020) systematic review.

Illueca & Doolittle study is recommended, since very specific personal prayer types were identified that seemed beneficial for various types of pain and a correction of the statements noted above is necessary.	
Reviewer: 1 **In addition, I would advise the authors to stay away from the grey literature, with the only exception of preprints, unless there is a clear rationale for an RCT to remain unpublished. The lack of peer review of any data included would diminish the objectivity and value of their findings.**	We agree with the reviewer that some grey literature might not have been submitted to peer-review, and that it is necessary to implement additional procedures to evaluate the quality of evidence of studies from grey literature. However, we respectfully disagree with the reviewer that the grey literature should be excluded from systematic reviews. By including the grey literature and evaluating the quality of all studies reviewed, we are following the current standards and recommendations of rigorous reviews (e.g., COCHRANE recommendations). Doing so also helps to minimize the effects of publication bias. Also, by evaluating the quality of the research using standardized procedures, we address the issue of research quality. See, for example, section 4.3.5. of the "Cochrane Handbook for Systematic Reviews of Interventions" relative to the inclusion of grey literature.
Reviewer: 1 **I also would recommend a revision of the nine original studies included by Illueca & Doolittle (2020).** **The introduction of the present protocol could thus be more complete by including the following two studies in Muslim patients reported in the literature:** **Rahman, F. S., Yahya, N., Din, N. M. M., Izaham, A., & Mat, W. R. W. (2018). The comparative effects of listening to prayer recitation and music therapy intraoperatively on postoperative pain. International Medical Journal Malaysia, 17(2), 107-115.** **Hassanpour-Dehkordi, A., Fatehi, D. & Solati, K. (2016). Analgesic plus prayer versus analgesic alone. effect of prayer on intensity of postoperative pain, anxiety and physiological indices in surgical patients. A randomized clinical trial. Heroin Addiction and Related Clinical Problems,18(6),13-20.**	We thank the reviewer for this suggestion. We revised the text in the Introduction accordingly.
Reviewer: 1 **Regarding the search strategy and definitions of variable:** **- Previous researchers (see above) have noted in their publications the challenges of**	We are following COCHRANE recommendations which are clear on the importance of casting a wide net for the initial search, and then having clear inclusion and exclusion criteria. This reduces

using very broad search terms and a strategy for streamlining the terms and also for detailing papers to be excluded needs additional clarification	the risk of missing articles.
Reviewer: 1 **- It is advisable to also consider the International Association for the Study of Pain's (IASP) definition of pain for consistent with published literature, as this protocol will explore the emotional and spiritual aspects of pain**	Thank you for this suggestion. We revised the text to cite IASP's definition of pain, and note that because it is a subjective experience, we limited our review to studies that included a self-report of pain as the primary outcome.
Reviewer: 1 **- Likewise, for consistency with previous systematic reviews, please clearly define the types of prayer to be studied and provide at least one clear definition of prayer, including what is meant by individual versus group prayers which are very distinct practices from each other as are meditation, yoga, kung fu, reiki, etc.**	Thank you for this suggestion. We revised the text to note the distinctions between different types of prayer early in the text (i.e., participatory vs. distant intercessory prayer, and individual vs. communal [dyadic vs. group-based] prayer), and we provide a clear definition of prayer as used in the current study.
Reviewer: 1 **3- Study design is not appropriate to answer the research question. Please see refer to the introductory comments above.** **The chosen area of study, prayer and pain is an area already explored in earlier spirituality and health research, and the authors may want to reconsider the focus their study in a more stringent way so as to build upon rather than repeat or revisit areas of research that have fallen into disfavor in the clinical research community.** **Therefore, the main flaw of the study design, which is simple to resolve, is the intent to mix two very different types of prayers as interventions which would not be appropriate to answer the research question in a way that would provide a meaningful interpretation of the data. Previous research has largely indicated the inability to properly control prayers used in the community separate from the person at the center of the prayer efforts. It would be a mistake to conflate distant prayer by third parties with personal prayer types and it would be a lost opportunity to provide meaningful data and obscure a potential real relationship between one type of well-defined prayer and the chosen pain-related outcomes.**	As noted in our response to previous recommendations, we revised the eligibility criteria to include only studies that examined the effects of participatory prayer (i.e., we do not include distant intercessory prayer studies), and make a distinction between these different prayer types.
Reviewer: 1	

6- Are the outcomes clearly defined? As stated above, it is highly improbable to be able to interpret data that is based on a mixture of prayer types. **- Research that uses "prayer" as a catch-all umbrella term representing a potential therapeutic intervention can no longer be approached in generic terms. Group based prayers (distant and intercessory) and personal prayer are as distinct as group therapy versus individual therapy. To consider "prayer" as a mixture of techniques and styles severely compromises an accurate interpretation of data, analogous to including both antibiotics and anti-hypertensive medications in the same study in order to prove that a pharmacological intervention can be accurately evaluated without attention to the type, dose and administration routes and frequency of those drugs.**	As noted previously, we revised the text to more clearly define outcomes, we propose to exclude intercessory prayer, and we propose to examine outcomes separately for different types of participatory prayer. We agree with these points and believe we have addressed them in the revised Introduction and Method.
Reviewer: 1 **- Based on existing research around distant intercessory prayer, there is now a consensus among researchers that the contradictory data on distant intercessory prayer does not justify further research.**	We agree with this point, revised the text to note it in the Introduction, and limit our review to participatory prayer (individual, dyadic, and group-based).
Reviewer: 1 **A major flaw of studies on group-based prayers is the inability to control it as a variable where different people pray in their own style for varying durations and with different levels of emotional involvement with the prayer intentions.**	We agree with this point, although this is also a limitation of all types of prayer – different people will likely pray for different durations and with varying levels of emotional involvement. This is a very important point that we did not emphasize enough in our original submission. We address this by planning to extract data, when possible, regarding: the length of prayer, the frequency of prayer, the level of engagement and emotional involvement of the study participants, the religious affiliation of the study participants, the type of religious practice of the participants (being both religious and a practitioner, being religious but not a practitioner, self-proclaimed as "spiritual" or not, etc.), and attitude toward the religion (negative attitude towards deity vs. positive attitude towards deity). We plan to discuss these issues in the planned Systematic Review manuscript as factors that could impact outcome, and that are therefore deserving of future study. For example, investigators could (and should) measure duration, emotional involvement, valence, and expectations, to determine how these relate to outcomes. The findings from this research would not only increase our understanding of the effects of prayer but could help make prayer interventions more effective.

Reviewer: 1 **6- Are the study limitations discussed adequately? Yes, but there is room for improvement based on the published literature.** **- Previous research has shown the high degree of heterogeneity of the data related to prayer. The literature shows that methodology of the studies of spirituality, prayer, health and pain is mostly inconsistent between studies described. This substantive limitation should be addressed.**	We revised the manuscript to make this point clearer in the Introduction. Since the present manuscript is a protocol of a Systematic Review, this is not a limitation of the present study (protocol of a systematic review) itself, but rather of the state-of-the-art relative to prayer-based interventions. We plan to discuss this very important point in our planned Systematic Review manuscript and highlight that this aspect needs to be addressed in future studies to improve research in this area.
Reviewer: 1 **- There is a serious lack of measurement tools and validated questionnaires to characterize prayer used by pain patients and most studies do not clearly define the meaning of prayer. There is an added level of heterogeneity by the use of varying types of scales, and the also the use of mixed methodology.**	We agree with the reviewer and revised the manuscript to make this point clearer in the Introduction. Per above, the present manuscript is a protocol of a Systematic Review. Thus, this is not a limitation of the current manuscript, but rather of the state-of-the-art in the field. We plan to discuss this in our planned Systematic Review manuscript and make this point very clear in the Discussion of our planned Systematic Review.
Reviewer: 1 **- Due to the reported lack of uniformity in research methods it is highly desirable that future study designs like the one proposed in this protocol really consider ways in which to provide consistency of definitions of variables especially those pertaining to prayer types.** **It is my opinion that by giving consideration to the focusing the choice of prayer type as an intervention, and avoiding the mixing of two different interventions as discussed above, this protocol could help expand the existing knowledge base around the use of prayer as a potential intervention in the management of pain.**	Thank you very much for your extremely helpful comments and suggestions. We agree that considering these issues – and revising the text to address them, as described above – has improved the paper considerably. We are very grateful for the thoughtful review.
Reviewer: 2 **Dr. Bonnie Huang Hall, University of California San Francisco** **I am not on expert on conducting systematic reviews, nor meta-analysis. I leave that for other experts to comment on.** **The question the paper tries to answer is the effect of prayer as an intervention for all types of pain. However, the title implies the question is whether prayer is effective for**	We agree with the reviewer. We have changed the title of the manuscript so that it now reads, "**Prayer as a pain intervention: Protocol of a systematic review of randomized controlled trials**".

pain management.	
Reviewer: 2 **It may help to have predefined notions of what is effective pain control - given the title implies that the question about to be answered is whether prayer is effective for pain control? Consequently, possible definitions for what is effectiveness (ie commonly accepted definitions of effectiveness) include 30% reduction in quantitative pain scores, very effective often is 50% reduction in pain scores.).**	Thank you for this suggestion. In addition to obtaining information regarding the effect size(s) of prayer on outcome (i.e., pain intensity), we now propose to obtain information regarding the rates of meaningful reductions in pain intensity – also known as responder analyses – when possible. Specifically, we will record the rates by which different groups obtained a 30% and a 50% reduction in pain intensity.
Reviewer: 2 **The data being gathered from each study may be too generic. Just simply "type of prayer" and "frequency of prayer". It may not be able to show conclusive results since the subject's attitude toward the religion and religious/spiritual beliefs itself may be factors that may influence whether pray is effective.** **Since the studies are randomized, it may be useful to have other data points added that characterize the subjects stance on religion such as what religious /spiritual affiliation the subjects had, or what type of practitioner they are of that religion/spiritual affiliation (practicing, no religious affiliation, negative attitude towards deity vs positive attitude towards deity, are they self-proclaimed as "spiritual" or not, etc.**	Thank you for noting these issues. This point was also addressed by another reviewer. We revised the Method to plan to extract data about the length of prayer, frequency of prayer, level of engagement and emotional involvement, religious affiliation, type of practitioner participants are of that religion (religious and practitioner, religious but not practitioner, self-proclaimed as "spiritual" or not) and attitude toward the religion (negative attitude towards deity vs positive attitude towards deity), and by planning to discuss in the planned Systematic Review manuscript these issues of individual variation as factors that could impact outcome.
Reviewer: 2 **Perhaps a clarification of "type of prayer" can be made. Do the authors refer to group prayer, individual prayer, asking for prayer, or does the author refer to active prayer vs passive prayer. i.e. Type of prayer can be broken down into separate data categories, such as who is praying, how are they praying, etc. it will be hard to reproduce the study if the parameters of "type of prayer" are not specified**	This point was also raised by another reviewer, and addressed as described above.
Reviewer: 3 **Dr. Alisa Johnson, University of Florida** **I commend you for taking on this topic and I look forward to reading about the study when it is completed. I have no suggestions for revisions. Well done!**	Author response: We thank the reviewer for her kind comments.

VERSION 2 – REVIEW

REVIEWER	Marta Illueca Berkeley Divinity School, Yale Program Medicine Spirituality and Religion
REVIEW RETURNED	28-May-2021

GENERAL COMMENTS	I commend the authors for their careful and thoughtful integration of previous constructive feedback into the study design. This study will build upon and augment the preliminary research so far reported in the medical literature. The focus of this study is an area of interest within the emerging specialty of spirituality and healthcare. The findings will contribute to much needed understanding of the role of prayer, an universal spiritual practice, and its potential therapeutic impact on the management of chronic pain, an unmet medical need. We commend the research team for this endeavor and we remain at their disposal for any consultative need in this effort.

REVIEWER	Bonnie Huang Hall University of California San Francisco
REVIEW RETURNED	26-May-2021

GENERAL COMMENTS	Thank you for characterizing prayer type more specifically. Also I believe your group added more detail concerning statistics. I am not familiar with this so please consult an expert. In addition, the English is pretty well written, however please be aware of the "to use" verb.. a few times in the article the past tense would have generally been regarded as more standard English. Good luck in your study, analysis, and I hope we see some conclusions on particatory prayer.

VERSION 2 – AUTHOR RESPONSE

REVIEWERS' COMMENTS	AUTHOR RESPONSE
Reviewer: 1 **I commend the authors for their careful and thoughtful integration of previous constructive feedback into the study design. This study will build upon and augment the preliminary research so far reported in the medical literature.** **The focus of this study is an area of interest within the emerging specialty of spirituality and healthcare. The findings will contribute to much needed understanding of the role of prayer, an universal spiritual practice, and its potential therapeutic impact on the management of chronic pain, an unmet medical need. We commend the research team for this endeavor and we remain at their disposal for any consultative need in this effort.**	We thank the reviewer for her kind comments, and for her willingness to act as a consultant of this study.
Reviewer: 2 **Thank you for characterizing prayer type more**	We thank the reviewer for her kind comment.

specifically. **Also I believe your group added more detail concerning statistics. I am not familiar with this so please consult an expert.**	We have consulted an expert when developing the protocol of this systematic review, to ensure that the metanalysis plan is correct. No further details concerning statistics were added in the revised version of the manuscript relative to the original submission. The only alteration in the revised version relative to the original submission in regard to the statistical analysis was as follows: "Outcomes from *private participatory* prayer and from *communal participatory* prayer will be analyzed separately." (cf. page 15, lines 15 and 16)
In addition, the English is pretty well written, however please be aware of the "to use" verb.. a few times in the article the past tense would have generally been regarded as more standard English. **Good luck in your study, analysis, and I hope we see some conclusions on particatory prayer.**	This alteration was introduced to comply with the suggestion and concerns raised by the reviewers in their review of the original submission. We thank the reviewer for bringing to our attention our incorrect use of the verb "to use" in the manuscript. We examined each use, and made a correction when needed. We thank the reviewer again for her careful review and feedback, which we think improved the protocol paper and will result in a better review.